Multi-Agent Reinforcement Learning for America's Cup Regattas

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Abstract

In this project, we explore the application of Multi-Agent Reinforcement Learning (MARL) to train sailboats to compete in a simulated America's Cup-style regatta. Using Proximal Policy Optimization (PPO) in a vectorized multi-agent environment, we developed agents capable of learning navigation strategies, managing foiling dynamics, and reacting to a stochastic wind field. The final policy achieves a high race completion rate in evaluation, while preserving realistic constraints such as gate crossings, boundary limits, and collision-related penalties.

1 Introduction

This section will introduce the context of autonomous sailing and the motivation for using Reinforcement Learning in competitive regatta scenarios. It will briefly describe the main challenges of the task: stochastic wind, continuous boat control, tactical interaction between boats, and the need to respect simplified racing rules.

2 Background

This section will present the theoretical foundations used in the project, including Markov Decision Processes, policy optimization, and Proximal Policy Optimization. It will also summarize the sailing concepts needed to understand the simulator, such as true wind angle, velocity made good, polar performance curves, tacking, gybing, and foiling.

3 Methodology

This section will describe the system architecture and the formulation of the learning problem. The discussion will include the simulated racecourse, the stochastic wind model, the boat physics model, the multi-agent environment, the observation space, the continuous action space, and the reward shaping strategy.

4 Experimental Setup and Results

This section will report the training configuration, PPO hyperparameters, evaluation protocol, and empirical results. It will include quantitative metrics such as completion rate, reward evolution, final distances, termination reasons, and qualitative observations from generated race videos.

5 Conclusions

This section will summarize the achieved results and discuss the main limitations of the current simulator. Possible future extensions include richer America's Cup rules, wind shadow effects, more accurate hydrodynamics, independent policies for the two boats, and larger multi-boat race scenarios.